

Coding & Solution

Date	15 March 2026
Team ID	VBCUA-01
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of the implemented VBCUA code and the overall solution. The submission demonstrates working AI-based concept analysis, clean code practices, and alignment with the defined project requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for the chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/srinancharaiah167/VBCUA
Programming Language(s)	Python
Framework(s) Used	Streamlit; OpenAI Whisper; Sentence-Transformers; Librosa; ReportLab
Key Features Implemented	Audio upload, speech-to-text transcription, semantic similarity, audio feature analysis, filler word detection, weighted scoring, waveform visualization, and PDF report download.
Pending / Incomplete Features	No major core feature pending. Future scope: database storage, teacher dashboard, login system, and history tracking.
Setup / Run Instructions	Create virtual environment, install requirements.txt, activate environment, and run: streamlit run app.py

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The project uses a modular Python structure. Core functions are separated into transcription, semantic analysis, audio analysis, filler detection, scoring, and report generation modules. The project does not use a database; runtime files are saved locally.